

LOKESH R M

Robotics Engineer · ROS 2, Legged Locomotion, Sim2Real, Edge AI

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SUMMARY

Robotics engineer at Sirena Technologies building biped, quadruped and service robots from simulation through hardware. Work covers MATLAB and Isaac Sim locomotion simulation, walk sequences and control algorithms, the physical robot controller, and ROS 2 Nav2 and SLAM navigation on NVIDIA Jetson. Lead the cloud and perception services behind 80+ concurrent robots.

EXPERIENCE

AI Engineer - Sirena Technologies, Bengaluru, India

Jul 2026 - Present

- Deploy ROS 2 Nav2 and SLAM navigation for autonomous service robots used in material transport, and for a 4 foot autonomous biped.
- Fine tuned YOLO and SSD object detection models for office and industrial environments, and optimise inference with ONNX for NVIDIA Jetson Orin Nano.
- Led a 5 person team building **Sirena Cloud**, the voice, vision and LLM service layer supporting 80+ concurrent robots on an NVIDIA RTX Pro 6000.
- Built the streaming voice pipeline behind human robot interaction, combining VAD, ASR, TTS and LLM reasoning at roughly 60 ms response latency.
- Architected **Agent OS**, a 53 service multi agent platform deployed to 3 enterprise clients, and own its Policy Gateway enforcing 153 policies. Platform team of 12.

Robotics Engineer, Graduate Trainee - Sirena Technologies, Bengaluru and Solan, India

May 2025 - Jun 2026

- Built MATLAB and Python locomotion simulations for biped and quadruped robots covering walk sequences and control algorithms, and validated every platform in simulation before hardware build.
- Developed features in the LS6 customised robot controller codebase and implemented ROS 2 motion control for robotic arms and wheeled robots.
- Integrated robot sensor suites and the state estimation algorithms behind them.
- Led a 3 person team building the cloud system for an ESP32 P4 based home robotics product, including ESP-IDF firmware integration.
- Completed the graduate engineering rotation at the Solan facility and converted to a full time engineering role.

Machine Learning Intern - MURTI Labs, GITAM Bengaluru, India

May 2024 - Apr 2025

- Built autonomous delivery and emergency response drone systems: trained LSTM based object detection and action recognition models on drone video to 83% accuracy, and implemented path mapping for navigation.
- Built the image processing and data collection pipeline feeding model training.

Freelance Full Stack and AI Engineer - Independent, Bengaluru, India

2024 - Present

- Delivered production web and AI applications for 5+ clients, covering LLM integration, agent workflows, APIs and backend infrastructure.

SKILLS

Robotics: ROS 2, Nav2, SLAM, Navigation, Path Planning, Motion Control, Control Algorithms, Robotic Arms, Wheeled and Legged Robots, Biped and Quadruped Locomotion, State Estimation, Sensor Integration

Simulation and Sim2Real: MATLAB, Gazebo, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, Webots

Perception: YOLO, SSD, OpenCV, MediaPipe, ONNX, PyTorch, TensorFlow, LSTM

Edge and Embedded: NVIDIA Jetson Orin Nano, RTX Pro 6000, ESP32 P4, ESP-IDF, Edge Inference, Linux

AI Systems: LLMs, Multi Agent Systems, ASR, TTS, Voice Activity Detection, RAG

Programming and Infrastructure: Python, C++, C, MATLAB, SQL, Docker, Kubernetes, FastAPI, Kafka, Redis, Google Cloud, PostgreSQL, Git

PROJECTS

Simulation to Real Transfer for Legged Robots, Sirena

2025 - 2026

Built MATLAB and Python locomotion simulations for biped and quadruped platforms covering walk sequences and control algorithms, validating every robot in simulation before hardware build and carrying the work through to physical controllers.

Stack: MATLAB, Python, Gazebo, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, ROS 2

EDUCATION

B.Tech, Computer Science and Engineering - GITAM (Deemed to be University), Bengaluru

Aug 2022 - Aug 2026

CGPA: 7.9 / 10

ACHIEVEMENTS

- **INSPIRE Award MANAK** (Government of India, state level) · **Young Scientist Award** (Government of Karnataka, district level) · **Rajya Puraskar** (Governor's Award) · Finalist, Internal Smart India Hackathon, GITAM
- **Secretary, IEEE Computer Society GITAM** (Dec 2024 - Aug 2025) · Student Placement Coordinator, GITAM Career Guidance Cell